both male and female reproductive organs in a single individual. e.g- Liverfluke, Tapeworm.

17.2.4 Phylum Aschelminthes (Round worms)

Aschelminthes are bilaterally symmetrical, triploblastic animals. The body cavity is a **pseudocoelom**. They exist as **free-living soil forms** or as **parasites**. The body is round and pointed at both the ends. It is unsegmented and covered by **thin cuticle**. Sexes are separate. The most common diseases caused by nematodes in human beings are **elephantiasis** and **ascariasis**. e.g-***Ascaris, Wuchereria***.



Male Female

Figure 17.6 Round worms

17.2.5 Phylum Annelida (Segmented worms)

These are bilaterally symmetrical, triploblastic, **first true coelomate animals** with **organ-system grade of organization**. Body is externally divided into segments called **metameres** joined by ring like structures called **annuli**. It is covered by moist thin cuticle. **Setae** and **parapodia** are locomotor organs. Sexes may be separate or united (hermaphrodites). e.g- ***Nereis, Earthworm, Leech***.



Earthworm Leech

Figure 17.7 Segmented worms

17.2.6 Phylum Arthropoda (Animals with jointed legs)

Arthropoda is the largest phylum of the animal kingdom. They are bilaterally symmetrical, triploblastic and coelomate animals. The body is divisible into head, thorax and abdomen. Each thorasic segment bears paired **jointed legs**. Exoskeleton is made of **chitin** and is shed periodically as the animal grows. The casting off and regrowing of exoskeleton is called **moulting**.

Body cavity is filled with **haemolymph** (blood). The blood does not flow in blood vessels and circulates throughout the body (open circulatory system). Respiration is through body surface, **gills** or **tracheae** (air tubes). Excretion occurs by **malphigian tubules** or **green glands**. Sexes are separate. E.g., Prawn, Crab, Cockroach, Millipede, Centipedes, Spider, Scorpion.



Centipede Millipede

Figure 17.8 Animals with jointed legs



Centipede means 'hundred legs'. But most species have only 30 pairs. Millipedes have two pairs of legs on each segment. This name means 'thousand legs'. But, most millipedes have only about a hundred.

Activity 1

Identify the following pictures of Arthropods.



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